Riko Research Report: The Whole Project

Oct 8, 2026 · @Dread

1. Executive Summary

Riko is three things: a small public voice-chat repo, a VTuber avatar layer shown only on video, and a paid app on a waitlist. Only the first can be checked in code, and its README overstates it.

Method and limits. For the repo, I cloned rayenfeng/riko_project and read every file, its 5 commits and all 28 issues and pull requests. I did not run it. For the avatar layer, I used the video research report on "My AI Waifu is A VTuber Now": 310 frames with OCR plus YouTube's auto-captions, since the file had no audio. For Rikoverse, I read its public landing page. I checked versions against GitHub, PyPI and OpenAI's docs.

Layer 1, the public repo: works, but the README is wrong in places. It is about 200 lines of Python: push-to-talk, Faster-Whisper, OpenAI `gpt-4.1-mini`, then a GPT-SoVITS server. The flow matches the code. But the config file name is wrong, the run command fails as written, the GPU steps do nothing (the code is fixed to CPU), and the bundled config points to the author's own `D:\` drive. It says "MIT" with no LICENSE file. Nothing has been committed since 2025-06-24.

Layer 2, the avatar: real technique, no public code. The video drives a VRoid model in VSeeFace with VMC/OSC messages from Python, and the calls it settles on match the VMC spec. ChatGPT's suggested `/VMC/Ext/Blend/Expression` address does not exist, and the video shows it failing. The motion is scripted idle and talking loops, not AI-driven. None of this code is in the repo.

Mismatched model story. The repo uses OpenAI's paid API. The video credits a local model, TheBloke/Ana-v1-m7-GGUF, which is licensed for non-commercial use only. The repo has no local-LLM option.

Layer 3, Rikoverse: a $7/month waitlist. The video's "secret link" leads to it. It promises memory and live avatar reactions; none of it can be checked. The repo, with 1,155 stars and 215 forks, never mentions it.

Outdated pieces. Python 3.10 (the tested version) reached end of life on 2026-10-01. The VSeeFace build shown lacks a Unity security patch. Claude 3.7 Sonnet, which wrote the avatar animations, was retired on 2026-02-19.

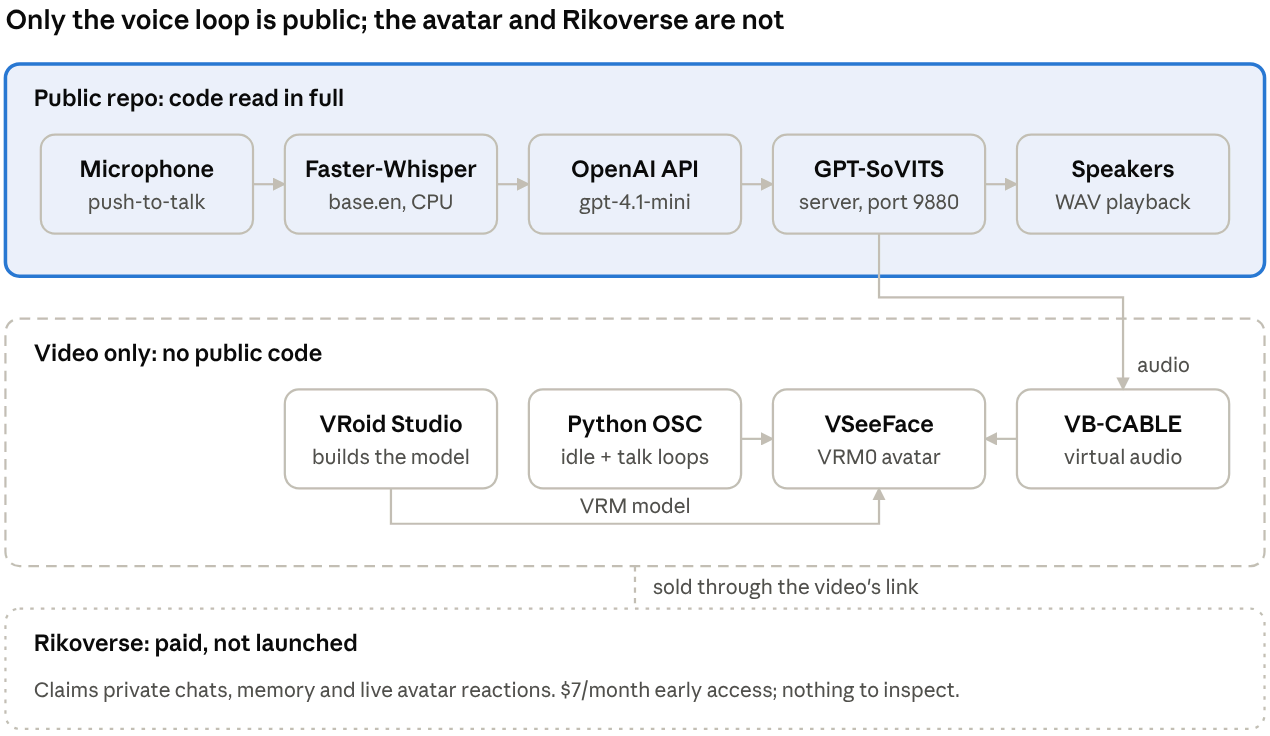
2. What Riko Is

Riko is an anime-styled "snarky" AI companion that listens, answers in a cloned voice and, on video, moves as a VTuber avatar.

Creator. "Just Rayen" (rayenfeng on GitHub), whose profile reads "Trying to become data scientist." He documents the build on YouTube and sells early access through Rikoverse.

The three layers:

Layer	What it does	Where it lives	Can it be checked?
Voice loop	Mic → Faster-Whisper → OpenAI `gpt-4.1-mini` → GPT-SoVITS → speakers, with chat memory	Public repo, `server/`	Yes, source read in full
Avatar	VRoid model in VSeeFace, moved by VMC/OSC messages from Python; lip sync from audio via VB-CABLE	Videos only	Partly, from on-screen code
Product	Private chats, memory, "live avatar reactions", evolving personality	rikoverse.com, $7/month waitlist	No, claims only



Riko's three layers · what is public, video-only and paid

The GitHub repo stops at the speakers. Everything that makes Riko a VTuber sits in the dashed layers, which you can only see on video or buy.

History. The repo was created and fully committed on 2025-06-24 (5 commits). Its description calls it "The public version of the Riko Project", implying a fuller private one. The VTuber video dates to 2025, going by the models it shows.

Files in the repo:

File	What it does
`server/main_chat.py`	The loop: record, transcribe, ask the LLM, speak, delete temp audio, repeat
`server/process/asr_func/asr_push_to_talk.py`	Records the mic between two Enter presses, then transcribes with Whisper `base.en`
`server/process/llm_funcs/llm_scr.py`	Calls OpenAI's Responses API and stores the whole chat in `chat_history.json`

File	What it does
`server/process/tts_func/sovits_ping.py`	POSTs the reply to GPT-SoVITS on `127.0.0.1:9880` and plays the WAV
`character_config.yaml`	API key, model name, system prompt, voice settings
`character_files/main_sample.wav`	The reference voice clip (735 KB)
`client/still_in_development.txt`	Empty placeholder

3. Key Claims

Of 27 claims checked across the repo, the video and Rikoverse, 7 are verified, 7 mostly accurate, 7 misleading, 3 false and 3 cannot be verified.

#	Layer	Where	Claim	Verdict	Evidence
1	Repo	README intro	"She listens, and remembers your conversations"	Mostly accurate	Full history is saved to `chat_history.json` and re-sent every turn. It is never trimmed.
2	Repo	README intro	A "fully configurable conversational pipeline"	Misleading	Only prompt, model and voice are in config. Whisper model, CPU device, server URL and sample rate are hard-coded.
3	Repo	README, Configuration	"All prompts and parameters are stored in `config.yaml`"	False	The file is `character_config.yaml`. No `config.yaml` exists.
4	Repo	README, top	"tested with python 3.10 Windows >10 and Linux Ubuntu"	Cannot be verified	No tests or CI. Python 3.10 reached end of life on 2026-10-01.
5	Repo	README, Setup	Install with `uv pip install -r extra-req.txt` then `requirements.txt`	Mostly accurate	Installs, but `requirements.txt` is mostly GPT-SoVITS's dependencies. `install_reqs.sh` uses steps the README never mentions.
6	Repo	README, Setup	GPU support for Faster-Whisper needs CUDA and cuDNN	Misleading	`main_chat.py` hard-codes `device="cpu"`.
7	Repo	README, Usage	"Launch the GPT-SoVITS API"	Mostly accurate	The `/tts` call on port 9880 matches GPT-SoVITS's `api_v2.py`. No launch steps given.
8	Repo	README, Usage	Run `python main_chat.py`	False	The script is `server/main_chat.py`. It works only as `python server/main_chat.py` from the repo root.
9	Repo	README, Usage	Six-step flow: listen, transcribe, GPT, respond, synthesize, play	Verified	Matches `main_chat.py` line for line.

#	Layer	Where	Claim	Verdict	Evidence
10	Repo	README, Credits	GPT-SoVITS, Faster-Whisper, OpenAI GPT	Verified	All three are used. The first two are MIT licensed.
11	Repo	README, License	"MIT — feel free to clone, modify"	Misleading	No LICENSE file. GitHub reports no license.
12	Repo	README, TODO	GUI, live mic, emotion control and VRM frontend are future work	Verified	None exist. No commits since 2025-06-24.
13	Repo	Description	"The public version of the Riko Project"	Misleading	Implies a usable cut of what the videos show. No avatar, client or local-LLM code.
14	Repo	Config default	Works with the bundled voice sample	Misleading	`ref_audio_path` points to `D:\PyProjects\waifu_project\...`, the author's own drive.
15	Video	Title, 0:00	"My AI Waifu is a VTuber now"	Mostly accurate	Motion is scripted idle and talking loops plus audio lip sync, not driven by the LLM.
16	Video	2:02	ChatGPT: VMC beats "OSC Expressions" (`/VMC/Ext/Blend/Expression`)	False	No such address in the VMC spec, and VMC itself runs over OSC. The video shows it failing (2:35–3:32).
17	Video	~4:02	VSeeFace's VMC receiver defaults to port 39539	Verified	VMC spec and VSeeFace docs.
18	Video	4:21–4:47	Blink via `Blend/Val` then `Blend/Apply`	Verified	Matches the spec. Names are case-sensitive.
19	Video	4:50–5:10	`Bone/Pos` numbers 4–6 are nod, twist and side-bend	Mostly accurate	They are quaternion x, y, z, which act like pitch, yaw, roll only for small angles.
20	Video	~4:13	Joints are sent "every millisecond or something"	Mostly accurate	Spec says every update cycle. "Every millisecond" is hyperbole.
21	Video	5:11–7:22	Commands run slowly unless sent twice	Cannot be verified	Cause not shown. VSeeFace's smoothing slider is one guess.
22	Video	~8:05–8:17	"LLM leaderboards" show Claude is the better programmer	Misleading	The OpenRouter page ranks by token usage, not quality.
23	Video	8:18–8:58	The final animations came from Claude 3.7 Sonnet	Verified	On-screen chat and notebook agree.
24	Video	~9:00	GPT-SoVITS → WAV → VB-CABLE → VSeeFace	Verified	Consistent with each tool's docs.
25	Video	Credits	Local LLM is TheBloke/Ana-v1-m7-GGUF	Mostly accurate	The model exists, but is CC-BY-NC-4.0, and the repo uses OpenAI instead.
26	Rikoverse	Video description	"Secret link?? not virus"	Misleading	It is a paid early-access waitlist (`utm_campaign=alpha`).
27	Rikoverse	Landing page	Memory, "live avatar reactions", evolving personality, $7/month	Cannot be verified	Not launched. "Access opens soon"; nothing to test.

4. Detailed Fact Check

Each item separates what Riko's README or video says from what the code and independent sources show. Links are in section 7.

A. Repo and setup

A1. Config file and run command: False. The README says to edit `config.yaml` and run `python main_chat.py`. The real file is `character_config.yaml`, and the script is in `server/`. Both `llm_scr.py` and `sovits_ping.py` open `character_config.yaml` relative to the current folder. So the only working command is `python server/main_chat.py` from the repo root. Issue #21 ("files not found running the setup") is a likely result.

A2. Dependencies: Mostly accurate. `extra-req.txt` holds what Riko's own code uses: `faster-whisper`, `sounddevice`, `openai`. `requirements.txt` is about 45 packages, including Chinese, Japanese and Korean text tools (`funasr`, `pypinyin`, `pyopenjtalk`, `g2pk2`). Riko's code imports none of them; they match GPT-SoVITS's needs. `install_reqs.sh` takes a different path: CUDA 12.6 PyTorch 2.6.0, `--no-deps`, and NLTK downloads. The README never mentions it. `llm_scr.py` imports `gradio` but never uses it.

A3. GPU support: Misleading. The README lists CUDA and cuDNN "for Faster-Whisper GPU support". `main_chat.py` loads `WhisperModel("base.en", device="cpu", compute_type="float32")`. A GPU only helps GPT-SoVITS, which runs separately. Issue #24 asks about AMD GPUs; no maintainer answered.

A4. Voice sample path: Misleading. `ref_audio_path` is `D:\PyProjects\waifu_project\riko_project\character_files\main_sample.wav`. GPT-SoVITS reads that path itself, so anyone else must edit it first. The README does not say so.

A5. License: Misleading. The README says "MIT", but the repo has no LICENSE file and GitHub's API reports `license: null`. The intent is clear, but without the license text, reuse terms are informal. Who owns the voice in `main_sample.wav` is not stated.

A6. Maintenance: Verified as stale. All 5 commits are from 2025-06-24. 22 issues and PRs are open, including fixes (#12 "Fix crash on tts failure") and Ollama support (#16). The owner has not commented on any issue or merged any outside PR.

B. Voice and LLM pipeline

B1. Flow: Verified. `main_chat.py` runs exactly the README's six steps, then deletes every `.wav` in `audio/`.

B2. Memory: Mostly accurate. Every turn is appended to `chat_history.json` and the whole file is sent each time. `gpt-4.1-mini` has a context window of about 1M tokens, so it will hold a long time. But input cost grows every turn, and nothing summarizes or trims.

B3. Push-to-talk limits: Not mentioned. Recording runs between two Enter presses and is capped at 60 seconds (`sd.rec(int(60 * samplerate))`). Longer speech is cut silently.

B4. GPT-SoVITS call: Verified. `POST http://127.0.0.1:9880/tts` with `text`, `text_lang`, `ref_audio_path`, `prompt_text`, `prompt_lang` matches GPT-SoVITS's `api_v2.py`, whose default port is 9880. If the server is down, `sovits_gen` returns `None` but the loop still tries to play the file and crashes. A comment mentions a "START ALL SERVER SCRIPT" that is not in the repo.

B5. Model mismatch: Mostly accurate. The video credits TheBloke/Ana-v1-m7-GGUF, a local model under CC-BY-NC-4.0. The repo only calls OpenAI. Five issues (#4, #9, #13, #15, #20) ask for a free or local option.

C. Avatar control (VMC/OSC, video only)

C1. ChatGPT's "OSC Expressions": False. VMC is built on OSC. It defines `/VMC/Ext/Blend/Val` (name and value) and `/VMC/Ext/Blend/Apply`. There is no `/VMC/Ext/Blend/Expression`. The video honestly shows it failing (2:35–3:32).

C2. Blink and expressions: Verified. The `Blend/Val` + `Blend/Apply` pattern is what the spec describes. "Blink" is a VRM0 preset name; names are case-sensitive.

C3. Decoding `Bone/Pos`: Mostly accurate. The spec format is name, px, py, pz, qx, qy, qz, qw: a position, then a quaternion. Numbers 4–6 act like nod, twist and side-bend only for small angles, and the 7th (qw) is needed too.

C4. "Send twice" and "every millisecond": Cannot be verified / hyperbole. The spec says bones are sent each update cycle, at a rate set by the sender. The slow-command workaround is unexplained.

C5. Audio pipeline: Verified. GPT-SoVITS writes a WAV, VB-CABLE forwards it, and VSeeFace lip-syncs to that audio input. VB-CABLE is donationware; its commercial terms are unverified.

D. AI coding claims

D1. "Leaderboards say Claude is the better programmer": Misleading. The OpenRouter page shown ranks by token usage, not quality. Anthropic did claim state-of-the-art on SWE-bench Verified for Claude 3.7 Sonnet (2025-02-24), but the video does not cite it. That page now returns a 404.

D2. "The animations came from Claude": Verified as a statement of what he did, not of which model is better. Claude 3.7 Sonnet was retired on 2026-02-19.

E. Licensing and commercial

E1. The "secret link": Misleading. It goes to rikoverse.com with a campaign tag. The page is a limited-alpha waitlist: "Early Access $7/month", "No charge today", "Access opens soon". The sales motive is not said on camera.

E2. What Rikoverse promises: Cannot be verified. Private chats, memory, live avatar reactions and an evolving personality. The public repo has the memory part only, and no avatar code. The site does not mention open source or GitHub.

E3. Credited model license: Mostly accurate. Ana-v1-m7 is non-commercial. Whether the paid service uses it is unknown.

5. Important Information Riko Leaves Out

The biggest gaps are the missing avatar code, the API-key and network risks, and the real cost.

1. **The avatar code is not public.** The VMC/OSC scripts, notebooks and animations from the video are not in the repo. Issue #10 asks how the avatar works; only other users answered.
2. **API key in a tracked file.** The OpenAI key goes in plain text in `character_config.yaml`, which is committed, and there is no `.gitignore`. Pushing a fork can leak the key. `chat_history.json` (every conversation) is not ignored either.
3. **Listener bound to every interface.** In the video, the OSC listener binds to `0.0.0.0`. Anyone copying it should use `127.0.0.1`.
4. **Ongoing cost.** OpenAI is paid per token: `gpt-4.1-mini` is $0.40 per 1M input and $1.60 per 1M output tokens. Because the full history is re-sent every turn, each reply costs more than the last. Rikoverse adds $7/month.

5. **Hardware.** GPT-SoVITS is built around NVIDIA CUDA; `install_reqs.sh` pins CUDA 12.6. VSeeFace is Windows-only and supports VRM0 models only.
6. **No error handling.** If GPT-SoVITS is down, the loop crashes. If the OpenAI call fails, the turn is lost.
7. **Port setup.** Neither the repo nor the video shows GPT-SoVITS's launch command or VSeeFace's send and receive ports. VSeeFace recommends different ports for each so it does not talk to itself.
8. **Licensing.** No LICENSE file, a non-commercial model credited in the video, unstated rights to the sample voice, and unverified VB-CABLE terms.
9. **Simpler routes.** The mouth can be driven with the A/I/U/E/O blend shapes over `Blend/Val`, with no virtual cable. @pixiv/three-vrm renders VRM in a browser with no VSeeFace.
10. **Check LLM-written code.** The video's own lesson: ChatGPT's first OSC address did not exist. Check generated API calls against the spec.

6. Current Status

As of 2026-10-08, the Python version, the VSeeFace build, the Claude model and the OpenRouter page used for Riko have all been superseded. The repo has not changed since 2025-06-24.

Item	In Riko	Now
Repo activity	5 commits, all 2025-06-24	No commits since. 1,155 stars, 215 forks, 22 open issues and PRs, no owner replies
Python	"tested with python 3.10"	3.10 reached end of life on 2026-10-01 (final release 3.10.22)
OpenAI model	`gpt-4.1-mini`	Not on OpenAI's deprecations page. `gpt-4.1-nano` shuts down 2026-10-23
`openai` package	Unpinned	3.26.1

Item	In Riko	Now
PyTorch	2.6.0 with CUDA 12.6 (`install_reqs.sh`)	2.14.1
Faster-Whisper	Unpinned, `base.en` on CPU	v1.2.1 (2025-10-31), MIT
GPT-SoVITS	API v2 on port 9880	Latest release 20250606v2pro; repo still active (pushed 2026-10-07), MIT
VSeeFace	v1.13.38c β (video)	v1.13.38c5. c4 adds Unity's patch for CVE-2025-59489 (CVSS 8.4)
python-osc	Version not shown (video)	1.10.2, needs Python 3.10+
Claude 3.7 Sonnet	Wrote the avatar animations	Retired 2026-02-19
OpenRouter `/rankings/programming`	Used to pick Claude	Returns 404. `/rankings` ranks by token usage
Ana-v1-m7-GGUF	Credited local LLM (video)	Still CC-BY-NC-4.0
Rikoverse	"Secret link"	Limited alpha, $7/month early access, "Access opens soon"

7. Best Sources

Primary and official sources only. Issue threads are used for user experience, never for a verdict.

Checked for this report:

- rayenfeng/riko_project (code, commits, metadata)
- riko_project issues and pull requests
- Rikoverse
- OpenAI model deprecations
- OpenAI gpt-4.1-mini model page

- GPT-SoVITS and its api_v2.py
- Faster-Whisper and on PyPI
- openai on PyPI, torch on PyPI

Carried over from the video research report:

- VMC protocol specification
- VSeeFace official site and releases
- CVE-2025-59489 (NVD)
- Python 3.10 schedule (PEP 619)
- python-osc on PyPI
- Anthropic: Claude 3.7 Sonnet announcement
- Anthropic model deprecations
- OpenRouter rankings
- TheBloke/Ana-v1-m7-GGUF
- VB-CABLE
- three-vrm

8. What I Should Actually Believe

Believe the code you can read and the failures shown on camera. Do not believe the README's setup steps as written, or that the public repo is the Riko you see in the videos.

Definitely true

- The public repo is a working push-to-talk loop: Faster-Whisper, OpenAI `gpt-4.1-mini`, GPT-SoVITS, with full chat history saved to disk.
- `Blend/Val` + `Blend/Apply` and `Bone/Pos` are real VMC messages, and the video's final code uses them correctly.
- ChatGPT's `/VMC/Ext/Blend/Expression` does not exist.
- The avatar's motion is scripted. Claude 3.7 Sonnet wrote the final idle and talking animations.
- The repo has no avatar, VRM, client or local-LLM code, and no commits since 2025-06-24.
- Rikoverse is a paid waitlist at $7/month.

Probably true

- The repo runs once you fix the config path, run it from the root, and start GPT-SoVITS first.
- The audio-to-avatar pipeline (GPT-SoVITS → VB-CABLE → VSeeFace) works as shown.
- A "private version" exists, given the repo description and the video.

Questionable

- That the repo is "fully configurable" or uses the GPU for speech recognition.
- That "MIT" applies, with no LICENSE file.
- That the leaderboard shows Claude is "better". It shows usage.
- That "send it twice" is a real fix.

False

- That `config.yaml` exists or `python main_chat.py` works as written.
- That an OSC "Expressions" address is an alternative to VMC.

Cannot be verified

- Anything about Rikoverse beyond its landing page: features, model, whether it shares code with the repo.
- Whether the paid service uses the non-commercial Ana model.
- That the repo was tested on Windows and Ubuntu.
- Who owns the voice in `main_sample.wav`.

9. Practical Next Steps

To run Riko today, fix the config and run command, keep your API key out of git, use a supported Python, and build the avatar layer yourself.

Run the voice loop

1. Use Python 3.11 or newer (3.10 is end of life) in a fresh virtual environment.
2. Install and start GPT-SoVITS first: `python api_v2.py -a 127.0.0.1 -p 9880` from its own folder. An NVIDIA GPU is strongly advised.
3. In Riko, install `extra-req.txt`. You only need `requirements.txt` if GPT-SoVITS shares the same environment.
4. Edit `character_config.yaml`: set your OpenAI key and change `ref_audio_path` to the full path of `character_files/main_sample.wav` on your machine.

5. Add a `.gitignore` with `character_config.yaml`, `chat_history.json` and `audio/` before you commit anything. Better still, read the key from an environment variable.
6. Run `python server/main_chat.py` from the repo root. Press Enter to start and stop recording; keep it under 60 seconds.
7. For GPU speech recognition, change `device="cpu"` to `"cuda"` in `main_chat.py`.
8. To avoid OpenAI costs, look at the open Ollama pull request (#16) or point the OpenAI client at a local server. Check any local model's license first.

Add the avatar (not in the repo)

1. Install VSeeFace v1.13.38c5 or newer (Windows only). Export your VRoid model as VRM0.
2. Use `python-osc` to send `/VMC/Ext/Blend/Val` + `/VMC/Ext/Blend/Apply` for expressions and `/VMC/Ext/Bone/Pos` (all 7 numbers) for bones, to VSeeFace's receiver port (default 39539).
3. Bind any listener to `127.0.0.1`, not `0.0.0.0`, and use different send and receive ports.
4. For lip sync, route GPT-SoVITS audio through VB-CABLE, or send the A/I/U/E/O blend shapes directly.
5. Check every LLM-written OSC address against the VMC spec.

Before paying for Rikoverse

1. Ask what it contains and which model it runs. The public repo has no avatar code, and the feature claims are unverified.
2. Compare against the free path: everything above is free except OpenAI tokens and an optional VB-CABLE donation.